

NETS FOR ELECTRONICS DESIGN: Migrating From Amateur To Professional Designer

Nets are transforming electronics design! From messy wire connections to organised circuits, learn why professionals are making the switch

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Schematic circuit design is a key step for any electronics product design, which we later convert to PCBs and then generate the Gerber files for PCB production. However, the circuit diagrams and designs we often see during our engineering days look something like Fig. 1, with wires to show connections. Yet, when you transition to the professional world and delve into complex designs, you encounter designs like the one in Fig. 2. Most professional circuit designs and reference designs often resemble this. Many engineers, especially those who are self-taught or lack formal training in electronics design, may not be aware of this design, which uses nets.

For basic circuit design, the first option is acceptable. However, for professional product design and prototyping, if you need to make changes to the design and replace

any ICs, or diodes, it becomes a complex problem. Making such changes in the initial design, where we have wire lines to connect the pins and components on the circuit, can be challenging. If your design has many components, a mesh of wire lines makes it complex to read the design. It also becomes impossible to change or replace the ICs and components in the design. Many newbies and amateur design engineers unknowingly make this mistake and find themselves in a difficult situation when their design requires modification or updating.

This article is for those design engineers, especially those new to electronics design, who may have entered electronics design as hobbyists and learned through experience rather than pursuing a certification course in circuit design. Recently,

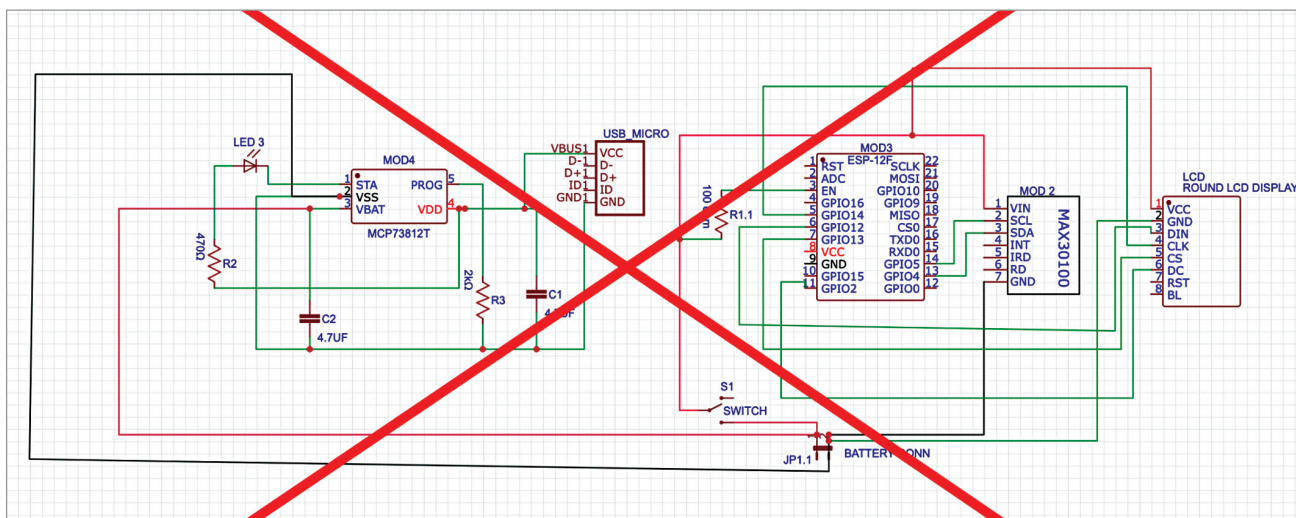


Fig. 1: Circuit design using normal wiring lines